Cyprus University of Technology



Computer Engineering & Informatics

English for Computer & Informatic Engineers II (LCE 116.1)

Our experience in 3D printing: Binary Clock

Student Names:

Andriana Georgiou - (33503)

Georgios Karyos - (30738)

Dimitris Chambi - (30727)

Lecturer:

Dimitrios Boglou

Word Count: **730**

We declare that:

a) this assignment is entirely our own work.

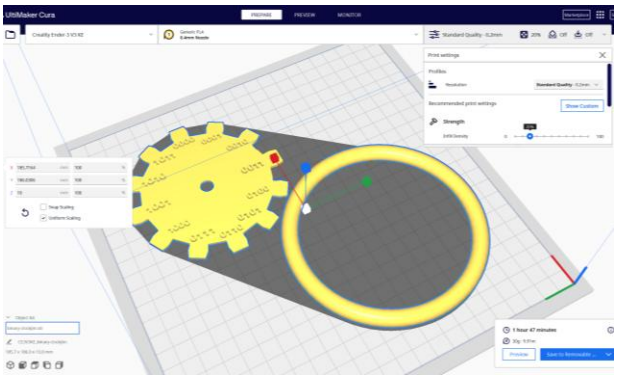
b) the material contained in this assignment has not previously been submitted for any other subject at the University.

March 7, 2025

Working with Ultimaker Cura and printing a 3D model was both a great and quite thrilling experience. Our idea was to create a binary clock where the numbers are based on the binary system. Our primary areas of attention were the technical settings of the Ultimaker Cura for 3D and the challenges we faced, the solutions we implemented. By the end, we had not only developed a functional 3D-printed model but also gained valuable insights into slicing software, troubleshooting, and teamwork.

Our team created a binary clock as an interesting challenge and a way to blend 3D printing with our field of study. The clock represents time in binary form, making it unique. The design required precise measurements and careful planning to ensure a smooth fit. Once complete, we imported it into Ultimaker Cura for slicing and adjusted the settings for optimal print quality. After slicing, we estimated the total print time to be about 1.5 hours and prepared the printer accordingly.

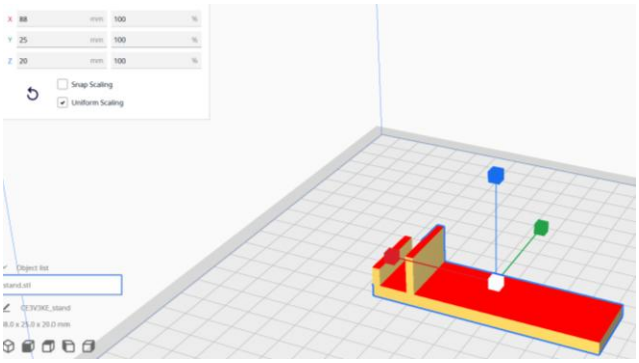
To manufacture our binary clock, we used Ultimaker Cura for slicing and optimizing 3D print settings. This ensured structural integrity while minimizing material usage and print time. The model was uniformly scaled to fit within the printable area, with final dimensions of 185.71 mm (x-axis), 186.03 mm (y-axis), and 10 mm (z-axis). The two parts combine into one file, preserving proportions while remaining practical for printing. For the print profile, we



selected Fast-Draft, balancing speed and detail quality. The 0.2mm layer height allowed for a smooth surface finish while keeping print time reasonable. To optimize material use and strength, we set the infill

density to 30% with a grid pattern, providing structural stability without unnecessary weight or waste. For durability, we set the shell thickness to 0.8 mm, ensuring strong outer walls without excessive filament use. The wall line count was adjusted to maintain rigidity while keeping material consumption efficient.

To support the binary clock, we designed and printed a custom stand using a Creality Ender-3 V3 KE with the help of a Mechanical Engineer. Measuring 88 × 25 × 20 mm, the stand provided a stable base while maintaining a sleek, minimal profile. We used 20% infill with a 0.2mm layer height, balancing strength and efficiency. Printed in black PLA, the stand seamlessly complemented the clock, requiring minimal post-processing before use.



During the printing process, we encountered unexpected challenges that required troubleshooting and adjustments. Initially, we did not notice the printing surface was unclean, causing poor adhesion and a failed print. After realizing the issue, we restarted with a properly cleaned surface. Even after cleaning, the print still did not stick well, so the 3D printing supervisor recommended using UHU stick glue to improve adhesion. This method worked effectively, helping the material adhere properly. However, while the glue improved adhesion, the print failed again. In the end, we used a cleaning spray to thoroughly clean the surface before attempting another print, ensuring it was free from adhesive and filament residue. These



challenges taught us the importance of preparing the print surface, following best adhesion practices, and maintaining the 3D printer for consistent results. Although these setbacks cost extra time, they ultimately improved our understanding of troubleshooting in 3D printing.

Through this project, we gained hands-on experience with 3D modeling, slicing software, and troubleshooting 3D printing issues. By collaborating, we divided tasks efficiently and improved problem-solving skills. Additionally, we explored how Cura settings influence print quality and completion time. To enhance the design, we would test different infill patterns for added strength and refine support structures to minimize post-processing. We would also experiment with printing temperatures to optimize adhesion and overall quality.



Exploring alternative filament materials could improve durability and finish, making the clock more aesthetically appealing and functional. Future improvements may include adding LEDs or electronic components to make the binary clock fully operational. These enhancements would expand our knowledge beyond 3D printing, integrating electronics and coding into our project.

Overall, this project enhanced our technical skills, problem-solving abilities, and teamwork, making it a valuable learning experience. The challenges we faced deepened our understanding of 3D printing workflows, troubleshooting, and collaboration. We now have a strong foundation in Ultimaker Cura and are better prepared for future 3D printing projects.